

SECURITY INCIDENT INVESTIGATION REPORT

Brute Force Login Attempt Detection

Report ID	IR-2026-001	CONFIDENTIAL
Analyst	Michael Isiuwa	
Date	May 13, 2026	
Severity	HIGH	
Status	Closed - Test Environment	

1. Executive Summary

On May 13, 2026 at 15:05 UTC-4, a brute force detection rule in Splunk Enterprise triggered an alert identifying multiple failed login attempts against two accounts on the monitored endpoint DESKTOP-O4UQ38M. The detection rule identified 8 failed authentication attempts within a 60-second window for both the FakeUser and analyst-account accounts, exceeding the defined threshold of 3 attempts.

This event was conducted as a controlled simulation to validate the detection capability of the Splunk SIEM deployment. The alert fired as expected, confirming the rule is functioning correctly. No malicious activity was identified — this was a purposefully generated test event.

2. Detection Rule

2.1 Rule Configuration

Field	Value
Rule Name	Brute Force Detection - Multiple Failed Logins
Event Code	4625 - Windows Failed Logon
Log Source	WinEventLog:Security
Detection Logic	3+ failed login attempts from same account within 60 seconds
Alert Type	Real-time, Per-Result
Severity	High
Throttle	60 minutes per Account_Name
SIEM Platform	Splunk Enterprise 10.2.3

2.2 SPL Query

```
index=main EventCode=4625 source="WinEventLog:Security" | bucket _time span=60s | stats count by _time, Account_Name | where count >= 3 | sort - count
```

3. Incident Timeline

Time (UTC-4)	Event
15:04:00	PowerShell script executed to simulate brute force attack against FakeUser account
15:05:00	Splunk detects 8 failed login attempts for FakeUser within 60-second window — alert triggered
15:05:00	Splunk detects 8 failed login attempts for analyst-account (calling account) within same window
15:06:00	Alert reviewed in Splunk — confirmed as controlled test. No further action required.

4. Findings

4.1 Affected Accounts

- FakeUser — non-existent account used as attack target in simulation
- analyst-account — local administrator account logged as the calling process

4.2 Event Details

Both accounts recorded exactly 8 failed authentication attempts within the 60-second window between 15:05:00 and 15:06:00 on May 13, 2026. The events were sourced from WinEventLog:Security on host DESKTOP-O4UQ38M. EventType=0 indicates a standard logon failure.

4.3 False Positive Analysis

In a production environment, the following scenarios could generate false positives for this rule:

- A user mistyping their password 3+ times within 60 seconds
- An automated service account with an expired or changed password
- A legitimate scheduled task using stale credentials

Mitigation: The Account_Name suppression field and 60-minute throttle reduce noise from repeat triggers. In production, a whitelist of known service accounts would further reduce false positives.

5. Recommended Response Actions

If this alert were to fire in a production environment, the following response steps should be taken:

- Step 1 — Verify the account owner: Contact the user associated with the flagged account to confirm whether they were attempting to log in.
- Step 2 — Check source IP: Identify where the login attempts originated. Internal IP indicates a local issue; external IP may indicate an active attack.
- Step 3 — Review account activity: Search Splunk for EventCode=4624 (successful logon) for the same account to determine if the attacker eventually succeeded.
- Step 4 — Containment: If suspicious, temporarily lock the account and escalate to Tier 2 analyst.
- Step 5 — Document: Record all findings, actions taken, and resolution in the incident ticket.

6. Conclusion

The brute force detection rule performed as expected in this controlled test environment. The Splunk alert successfully identified the simulated attack within seconds of the threshold being met. This validates the detection capability of the SIEM deployment.

The rule is now live and monitoring in real-time. Future enhancements could include integration with an automated response playbook to lock accounts automatically upon alert trigger, and enrichment with threat intelligence feeds to cross-reference source IPs against known malicious actors.

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Classification: Confidential - Home Lab